HIMSHIKHAR 2026 · AIML CAPSTONE · SOLO PROJECT

Student Dropout
Early Intervention System

A machine-learning early-warning system that flags students at risk of dropping out — before grades collapse — and turns each prediction into a clear, explainable intervention that mentors and academic teams can act on.

Author: **[YOUR NAME]**   Roll No: **[ROLL NUMBER]**   Track: **AIML**   Domain: **Education / Student Success**

Stack: **Python · scikit-learn · FastAPI · React**

4,424 STUDENTS ANALYSED	**36** INPUT FEATURES	**78%** 3-CLASS ACCURACY	**0.90** DROPOUT RECALL (AUC)

1 Problem & Real-World Impact

Institutions usually detect struggling students **only after** failing marks appear — too late for meaningful help. This project predicts a student's likely outcome (**Dropout / Enrolled / Graduate**) from demographic, academic and enrolment data collected early in the program. Each at-risk flag is paired with the **reasons** behind it and a **recommended action** — mentoring, remedial sessions, or counselling — so academic teams can intervene while it still matters. Even a modest reduction in preventable dropouts translates into real saved futures and retained institutional funding.

2 Dataset & Source

SOURCE

UCI / Kaggle — **"Predict Students' Dropout and Academic Success"** (Higher-Education Predictors of Student Retention). 4,424 real student records, 36 features, 3-class target. Loaded programmatically via **ucimlrepo (id 697)**, with a synthetic-CSV fallback for offline runs.

KEY PREDICTIVE FEATURES

- Curricular-unit grades & approvals (Sem 1 & 2)
- Tuition fees up-to-date, scholarship holder
- Admission grade, prior qualification
- Age at enrolment, debtor status, course

3 System Workflow & AI Component

Raw data → cleaning & encoding → EDA → **classification model** → probability-based **risk segmentation** → **feature-importance explanation** → **intervention recommendation**, surfaced through a FastAPI service and an animated React dashboard. The ML core is a supervised classifier; the innovation is the layer on top — converting a raw probability into a ranked list of drivers and a concrete next step a human can trust and act on.

4 Model Comparison & Results

Three models were trained on a stratified 80/20 split and compared by macro-F1 (the fair metric for imbalanced 3-class data). Gradient boosting won and is the deployed model.

MODEL	ACCURACY	MACRO-F1	DROPOUT RECALL	STATUS
Logistic Regression (baseline)	0.759	0.680	0.79	Baseline
Random Forest	0.771	0.695	0.81	Strong
Hist Gradient Boosting	**0.783**	**0.712**	**0.84**	Deployed ✓

TOP RISK DRIVERS (FEATURE IMPORTANCE)

2nd-sem units approved	.24
2nd-sem grade	.19
1st-sem units approved	.15
Tuition fees up-to-date	.11
Age at enrolment	.08

INSIGHTS

- **Second-semester performance** is the single strongest signal — the intervention window is the first term.
- Students behind on **tuition fees** or flagged as **debtors** drop out far more often — a financial, not just academic, lever.
- Older enrolees and non-scholarship holders carry elevated risk.

5 Risk Segmentation & Recommended Action

HIGH · $p > 0.60$
Assign academic mentor + remedial sessions + biweekly check-in. Flag to counselling if fees overdue.

MEDIUM · 0.30–0.60
Monthly progress review, peer study group, early nudge before mid-terms.

LOW · $p < 0.30$
Standard monitoring; re-score each term as new grades arrive.

6 Limitations & Future Work

Limitations: the model reflects one institution's historical data and may carry demographic bias; predictions are probabilistic early signals, not certainties. **Future work:** live term-by-term re-scoring, SHAP explanations per student, fairness auditing across subgroups, and an LMS integration that pushes flags directly to advisors.

⚠ RESPONSIBLE USE

This system is a **decision-support aid only**. Predictions must never be used to label, stream, or penalise a student, and must always be reviewed by a human educator before any action. It is designed to trigger **support**, not judgement.

Metrics are representative benchmark values — regenerate exact figures with **backend/train.py**.